

Homework 3: SQL

Danny Topete - dtpe004

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Problem 1:

Given the following relational database:

employee (employee-name, street, city)

works (employee-name, company-name, salary)

company (company-name, city)

manages (employee-name, manager-name)

Write the following Queries in SQL:

1. Find the names of employees who work in a city different from the city where their company is located.

```
SELECT e.employee-name
FROM employee e, works w, company c
WHERE e.employee-name = w.employee-name
AND w.company-name = c.company-name
-- Below outputs employees who work in the same city as their company, so
we use NOT EXISTS to get those who work in a different city
WHERE NOT EXISTS (
    SELECT *
    FROM employee e2, works w2, company c2
    WHERE e2.employee-name = w2.employee-name
    AND w2.company-name = c2.company-name
    AND e2.employee-name = e.employee-name
    AND c2.city = c.city
)
```

2. Find the name of employees who earn less than the average salary of their company.

```
SELECT e.employee-name
FROM employee e, works w
WHERE e.employee-name = w.employee-name
AND w.salary < (
    SELECT AVG(salary)
    FROM works
    WHERE company-name = w.company-name
)
```

Problem 2: Write the following queries in SQL on database shown below.

EMPLOYEE								
FNAME	LNAME	<u>SSN</u>	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNUM

DEPARTMENT			
DNAME	<u>DNUMBER</u>	MGRSSN	MGRSTARTDATE

DEPT_LOCATIONS	
<u>DNUMBER</u>	<u>DLOCATION</u>

WORKS_ON		
<u>ESSN</u>	<u>PNUM</u>	HOURS

PROJECT			
PNAME	<u>PNUMBER</u>	PLOCATION	DNUM

DEPENDENT				
<u>ESSN</u>	<u>DEPENDENT_NAME</u>	SEX	BDATE	RELATIONSHIP

Figure 1: Database Schema for Problem 2

1. For departments with an average salary greater than \$50,000, list the department name and the average salary of employees in each department.

```
SELECT d.dname, AVG(w.salary) AS average_salary
FROM department d, employee e, works w
WHERE d.DNUMBER = e.DNUM
AND e.SSN = w.ESSN
GROUP BY d.dname
HAVING AVG(w.salary) > 50000
```

2. Find the first and last names of employees who do not work on any project.

```
SELECT e.fname, e.lname
FROM employee e
WHERE NOT EXISTS (
    SELECT *
    FROM project p, works_on w, department d
    WHERE e.SSN = w.ESSN
    AND d.dnumber = p.dnum
    AND p.pnumber = w.pnum
)
```

3. Retrieve the last names of employees who share a birthday with at least one of their dependents.

```
SELECT DISTINCT e.lname
FROM employee e, dependent d
WHERE e.SSN = d.ESSN
AND e.BDATE = d.BDATE
```

4. Find the first and last names of employees who work on every project in the “Houston” location.

```
SELECT e.fname, e.lname
FROM employee e
-- Below outputs employees who do not work on any project in Houston, so we
use NOT EXISTS to get those who work on every project in Houston
WHERE NOT EXISTS (
    SELECT *
    FROM project p, department d, works_on w
    WHERE p.pnumber = w.pnum
    AND w.ESSN = e.SSN
    AND d.dnumber = p.dnum
    AND d.dlocation = 'Houston'
)
```

5. For each department, list the department name and the number of employees who work in that department.

```
SELECT d.dname, COUNT(e.SSN) AS num_employees
FROM department d, employee e
WHERE d.DNUMBER = e.DNUM
GROUP BY d.dname
```

6. List the last names of all employees who have no dependents

```
-- No need for DISTINCT because E with no D only appear once
SELECT e.lname
FROM employee e
WHERE NOT EXISTS (
    SELECT *
    FROM dependent d
    WHERE e.SSN = d.ESSN
)
```

Problem 3: Consider the following relational university database:

STUDENT (Name, StudentNumber, Class, Major)

COURSE (CourseName, CourseNumber, CreditHours, Department)

PREREQUISITE (CourseNumber, PrerequisiteNumber)

SECTION (SectionIdentifier, CourseNumber, Semester, Year, Instructor)

GRADE_REPORT (StudentNumber, SectionIdentifier, Grade)

1. Retrieve the names and majors of all 'A' students (i.e., students who have a grade of 'A' in all their courses).

```
SELECT s.Name, s.Major
FROM STUDENT s
WHERE NOT EXISTS (
    SELECT *
    FROM GRADE_REPORT g
    WHERE g.StudentNumber = s.StudentNumber
    AND g.Grade <> 'A'
)
```

2. Retrieve the names and majors of all students who do not have a grade of 'A' in any of their courses

```
SELECT s.Name, s.Major
FROM STUDENT s
-- Removes all students with an 'A' grade in any course
WHERE NOT EXISTS (
    SELECT *
    FROM GRADE_REPORT g
    WHERE g.StudentNumber = s.StudentNumber
    AND g.Grade = 'A'
)
```